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HOME PLOTS APPS FIGURE FORMAT TOOLS

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Files

Exp1_TestCase2.m

Exp1_TestCase3.m

Exp1.m

SlotC_Exp2.m

SlotC_Exp2a.m

SlotC_Exp2b.m

SlotC_Exp2c.m

SlotC_Exp3.m

SlotC_Exp4a.m

SlotC_Exp4b.m

SlotC_Exp4c.m

Workspace

Name	Value	Size	Class
d	1×50 double	1×50	double
f	[900,3500,28...	1×3	double
Freq	[900,3500,28...	1×3	double
Loss	[91.5249,103...	1×3	double
PL1	1×50 double	1×50	double
PL2	1×50 double	1×50	double
PL3	1×50 double	1×50	double

SlotC_Exp2a.m

```
1 clc;
2 clear;
3 close all;
4
5 % Distance (km)
6 d = 0.1:0.1:5;
7
8 % Frequency (MHz)
9 f = [900 3500 28000];
10
11 % Free Space Path Loss Calculation
12 PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
13 PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
14 PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
15
```

Figure 1

Propagation Loss vs Distance

Loss (dB) vs Distance (km) for 900 MHz, 3500 MHz, and 28000 MHz.

Propagation Loss vs Frequency

Loss (dB) vs Frequency (MHz) for d = 0.1 km.

Command Window

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TITLE SUBTITLE X-LABEL Y-LABEL Z-LABEL LEGEND COLORBAR GRID X-GRID Y-GRID

LABELS AND ANNOTATIONS

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SlotC_Exp4c.m

Workspace

Name	Value	Size	Class
Alloc	[1,1,1,1,1,0,0,0]	1×8	double
Blocked	[0,0,0,0,0,1,1,1]	1×8	double
TotalSlots	5	1×1	double
Users	[1,2,3,4,5,6,7,8]	1×8	double

SlotC_Exp2c.m

```
1 clc;
2 clear;
3 close all;

4
5 % ----- System Parameters -----
6
7 % Total users
8 Users = 1:8;
9
10 % Available TDMA slots
11 TotalSlots = 5;
12
13 % Initialize slot allocation
14 Alloc = zeros(1,8);
15
```

Figure 1

TDMA Slot Allocation per User

TDMA Slot Utilization Analysis

Command Window

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```
1      1
2      1
3      1
4      1
5      1
6      0
7      0
8      0
```

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SlotC_Exp4a.m

SlotC_Exp4b.m

SlotC_Exp4c.m

Workspace

Name	Value	Size	Class
FrameTime	4.6150	1×1	double
SlotDurat...	0.5769	1×1	double
Slots	8	1×1	double
t	[0,0.5769,1.1...	1×9	double

SlotC_Exp2.m

```
1 clc;
2 clear;
3 close all;
4
5 % ----- System Parameters -----
6
7 % TDMA Frame Time (ms)
8 FrameTime = 4.615;
9
10 % Number of Time Slots
11 Slots = 8;
12
13 % Duration of Each Slot
14 SlotDuration = FrameTime / Slots;
15
```

Figure 1

TDMA Time Slot Structure

Continuous View of Slot Allocation

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